



7th Global AI Hackathon

CHALLENGE

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The AI Apprentice

Accelerating the world's digital operations

POWERED BY

IIElevenLabs

Goals and Motivation

Our most knowledgeable generation is retiring. More than 11,200 Americans turn 65 every day. In Germany, 12.9 million workers, almost 30% of the labor force, will pass retirement age by 2036. Worldwide, people aged 65 and over will more than double to 1.6 billion by 2050. They take decades of judgment about how the world's digital operations really run. To build the economy of the future, we must reinvent how that knowledge is captured, mapped and taught to the next generation, before it walks out of the door.

Meet Sabine. She is 57 and has run accounts payable at a machine builder near Stuttgart for 24 years. It is Thursday, 4:10 pm, two days before month-end close, with 60 invoices still open. Lena, 26, who started on Monday, is watching Sabine's screen over a video call. Sabine moves one invoice to another cost center without a word, holds a second because that supplier double-bills every December, and sends a third for a second approval because it comes from the Czech subsidiary. Lena catches half of it. The 2019 process document covers even less. Sabine retires in 18 months, and 24 years of judgment will leave with her.

Everything needed already exists. Browsers share screens, and vision models can read what changes on them. An ElevenLabs voice agent can stay quiet while someone types, then ask the one question that matters at the right pause. What is missing is the apprentice that watches how screen work is really done, asks why, maps the workflow with its guardrails and teaches the next Lena what Sabine knows.

That gap is what this challenge asks you to close.

Current Challenges

KNOWLEDGE LIVES IN HEADS	What makes experienced people good at their job was never written down. New hires learn it slowly, over a shoulder.
RECORDINGS SHOW WHAT, NOT WHY	Screen recordings, how-to videos and task-mining logs capture clicks. They cannot tell a deliberate judgment call from a habit or a mistake.
GUARDRAILS ARE INVISIBLE	Limits, exceptions and the moment to stop and ask are rarely written down. New hires learn them by breaking them.

Your Challenge

Build **the AI Apprentice**: a working end-to-end MVP that captures what an expert knows while they work, maps it into a workflow anyone can follow and teaches it to the next generation. An ElevenLabs voice agent watches the expert's screen, asks why at the right moments and runs a short debrief until it fully understands the process.

The conversation is the core. The expert shares their screen and does a real task. The voice agent stays quiet while they type and asks short spoken questions at natural pauses: why this step, what would change the decision, what they would never do.

The debrief closes the gaps. When the task is done, the apprentice asks about the exceptions it noticed, the rules it is unsure about and the cases it has not seen. Then it explains the process back until the expert says: yes, that is how it works.

Your use case, your interface. Invoicing is our running example, but any knowledge-driven desk work fits: insurance claims, procurement, support escalations, KYC checks. The interface is yours too: a side panel, a floating voice companion, a replayable timeline or a coaching overlay. Experiment.

An apprentice, not a recorder

A recorder captures what happened. An automation tool copies the clicks. An apprentice asks why, learns the rules and guardrails behind each step, and keeps asking until nothing is unclear. If a new person could not do the task from what it learned, it is not an AI Apprentice.

Modules

Build all three.

1. Capture

A screen-share app with a voice agent that asks why while the expert works

2. Map

A debrief that turns the session into a clickable Work Map

3. Teach

A voice tutor that coaches a new hire on their own screen

Module 1: Capture

Build a web app where the expert shares their screen and an ElevenLabs agent listens in a side panel. Every second or two, send a frame to a vision model and turn what changed into events: invoice 4471 opened, cost center changed from 4711 to 0400. The agent stays quiet while the expert types, reads or talks, and asks at natural pauses: why this step, is there a limit, when would you stop and ask someone?

Required

During a real task, the agent asks at least three questions, each at a natural pause and about something visible on screen. At least one is about a guardrail.

Module 2: Map

When the task ends, the apprentice runs a short spoken debrief. It asks about what is still unclear, then explains the whole process back in its own words so the expert can confirm or correct it. The result is the Work Map: a clickable timeline where every step shows the screen moment, the decision, the reason in the expert's words and the guardrails around it.

Example step	Step 4 of 7: code the invoice to a cost center
Screen moment	03:12, invoice 4471, cost center field
Decision	Re-coded from opex (4711) to capex (0400)
Reason	"Equipment over €5,000 is always capex." Sabine, live question at 03:15
Guardrails	No asset number, no capex booking. Unknown supplier: stop and ask the controller.

Required	The debrief asks at least three follow-up questions that were not answered during the task and ends with a teach-back the expert confirms. Every step and guardrail links to a screen moment and the expert's own words.
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Module 3: Teach

Turn the Work Map into a voice tutor for the next generation. The new hire works a case on their own screen while the tutor watches, explains each step the way the expert did, asks them to predict the next decision and steps in before a guardrail is broken, replaying the expert's screen moment when it helps. At the end it shows what they have mastered and what to practice next.

Required	A judge playing a new hire processes a case the expert never showed. The tutor catches at least one wrong decision before it is saved and explains it using the reasoning the expert gave.
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The Apprentice Test

Timing is where most voice agents fail, but it is only the start. A good apprentice asks well, knows when it has understood and passes on what it learned. Your demo must answer five questions:

1. **When to ask.** How does the agent know the expert has paused, and stay quiet while they type, read or talk?
2. **What to ask.** How does it pick the question that reveals a reason or a guardrail, instead of one the screen already answers?
3. **When it has understood.** How does the debrief decide it is done, and how does the teach-back prove it?
4. **Whether the new hire learned.** How do you show they can handle a new case on their own?
5. **Trust.** How can the expert take something off the record, and how is personal data on screen protected?

Stretch Goals

We know that you can do it.

- **Two experts, one task.** Show where two sessions differ and ask each expert why.
- **Any language.** The expert explains in German; the tutor teaches a new hire in English.
- **Agent-ready guardrails.** Export the Work Map as instructions an agent can load, so it follows the same steps and stops where the expert would.

Think Bigger: The Moonshot

Your MVP captures one expert, one task and one new hire. We also want to see where it could go. End your pitch with one slide: the moonshot you would build next and how today's MVP gets you there. Some directions to start from:

- **A living company memory.** Every expert and every workflow in one map that stays current. When the work changes, the apprentice asks only about what is new.
- **The always-on apprentice.** No scheduled sessions. It notices a case it has never seen during everyday work and asks one question at the right moment.
- **People first, then agents.** The steps and guardrails that teach a new hire also let agents take routine steps safely, while people keep the judgment calls.
- **The world's operations manual.** Anonymized Work Maps across thousands of companies that show how the world's digital work is really done, and teach it to anyone, anywhere.

Built with ElevenLabs

The voice agent is the product, so it has to feel like a thoughtful colleague.

- **ElevenAgents** plays both roles, interviewer and tutor, with Expressive Mode for a curious, patient voice.
- **Your choice of LLM.** ElevenAgents lets you pick the model behind the agent. It decides what to ask, when, and when it has understood enough.
- **Scribe v2 Realtime** listens while people work and knows when they pause, so the agent speaks at the right moment.

One way to wire it

1. The browser shares the screen. A frame every one to two seconds goes to a vision model, which returns events, not video.
2. Client tools push those events into the ElevenAgents conversation, so the agent knows what is on screen.
3. After the task, an LLM merges events, transcript and answers into Work Map JSON and lists what is still unclear for the debrief.

4. The Work Map goes into the tutor's knowledge base and Procedures, and the tutor watches the new hire's screen the same way.

Two tips

- **Start with voice and one screen.** Get screen events into the agent's context before anything else.
- **Ask less, later.** Three to five live questions per ten minutes; the rest waits for the debrief.

Data Sources and Hints

Category	Source	Use
Your own workflow	Record yourself or a teammate	The most honest test data you will get
Agent brain	ElevenAgents LLM options: elevenlabs.io/docs/agents-platform/customization/llm	Choose the model behind the interviewer and the tutor
Screen understanding	Any vision-capable model, such as Claude, Gemini or GPT	Describe what changed between screen frames
Debrief and Work Map	An agent framework of your choice	Merge events, transcript and answers; find the gaps
Tools	ElevenLabs MCP tools: elevenlabs.io/docs/eleven-agents/customization/tools/mcp	Let the tutor look up guardrails through an MCP server
Tasks and roles	O*NET database: onetcenter.org/database.html	18,838 tasks across 1,016 occupations (CC BY 4.0)
Screen sandbox	WebArena: webarena.dev	Self-hosted web apps with fake data
Privacy	Microsoft Presidio: github.com/microsoft/presidio	Redact personal data in transcripts and frames

Bring Your Own Workflow

The best demo is a workflow you know well, from your own job or a teammate's. Run it on fake or sandbox data. A good one:

- Takes 5 to 10 minutes on screen.
- Hides at least one judgment call that is not written down anywhere.
- Has real guardrails: a limit, an exception, a moment to stop and ask someone.

No workflow of your own?

Use our running example: three supplier invoices in a sandbox ERP, one over the €5,000 capex line and one from a supplier that double-bills in December.

What Good Looks Like

A judge playing Sabine shares their screen and processes three invoices while talking. At a pause, a calm voice asks: "You moved that one to capex. What made you do that?" The judge explains that equipment over €5,000 is always capex. In the debrief it asks: "You held the December invoice. Is that for every supplier, and who decides when to release it?" Then it explains the whole process back in under a minute, and the judge corrects one detail. The Work Map shows seven steps, three judgment calls and four guardrails, each linked to its moment on screen. Then a second judge, playing a new hire, opens a fresh €7,200 equipment invoice and reaches for the opex code. The tutor says: "Sabine would stop here. Why do you think?" It replays her screen moment and lets them fix it. That is the bar.

What Makes a Strong Submission

Strong	Weak
Asks at natural pauses, about what is on screen	Interrupts mid-typing, or asks generic questions
Captures guardrails: limits, exceptions, when to stop and ask	Captures only the happy path
The debrief closes gaps and ends with a teach-back	A summary written from the transcript afterwards
The tutor teaches a new hire to decide, in the expert's words	A screen recording nobody will watch
A pitch with a clear moonshot and a path to it	A demo that stops at the demo

Why This Matters

The US Department of Labor's O*NET database lists 18,838 tasks across 1,016 occupations, and the judgment inside each rarely makes it onto paper. Gartner expects over 40% of agentic AI projects to

be canceled by the end of 2027, citing escalating costs, unclear business value or inadequate risk controls. Much of what those projects miss is knowledge that lives only in someone's head.

Voice changes who can share that knowledge: an expert who would never write a process document will happily explain their work while doing it. The company that gets this right turns every retiring expert into a Work Map and a tutor instead of a farewell party.

Appendix

- ElevenAgents quickstart: elevenlabs.io/docs/eleven-agents/quickstart
- ElevenLabs Startup Grants: elevenlabs.io/startup-grants
- Sources: UN News, Jan 2023 (news.un.org/en/story/2023/01/1132392); Alliance for Lifetime Income, Aug 2024 (protectedincome.org/news/analysis-peak-65-boom); Destatis, Aug 2022 (destatis.de/EN/Press/2022/08/PE22_330_13.html); Gartner, 25 June 2025 (gartner.com/en/newsroom); O*NET 31.0 Database (onetcenter.org/database.html)